

EXPERIMENT REPORT

Experiment 2 — Reinforcement Learning: Q-Learning

4×4 Grid World Environment

1. Introduction

Reinforcement Learning is a learning approach in which an agent interacts with an environment and improves its behaviour using rewards and penalties. Q-Learning is a model-free, off-policy algorithm that learns the expected value of actions in states.

2. Objective

Implement Q-Learning for a 4×4 Grid World, monitor Q-values during training, derive the learned policy and analyze cumulative reward convergence.

3. Environment Design

The grid has 16 states. State 0 is the start, state 15 is the goal, and states 5 and 10 are obstacles. The four available actions are Up, Down, Left and Right.

4. Parameters

$\alpha = 0.10$, $\gamma = 0.90$, $\epsilon = 0.20$ and 100 training episodes. Q-values are initially zero.

5. Implementation

At every step, the agent selects an action using ϵ -greedy exploration, receives a reward, observes the next state, and updates the corresponding Q-value using the Q-Learning update rule.

6. Q-Value Analysis

The experiment records complete Q-table snapshots after Episodes 1, 50 and 100. The visual comparison shows the transition from an uninformative zero table to values that encode useful directions toward the goal.

7. Final Policy

The final action at each state is selected using the highest Q-value. Obstacles are marked X and the goal is marked G. The resulting policy is stored in `q_learning_final_policy.csv`.

8. Reward and Convergence

Cumulative reward tends to improve as the agent learns. Remaining fluctuations are expected because ϵ -greedy exploration continues throughout training. A successful policy balances the +10 goal reward against repeated -1 step costs and -5 obstacle penalties.

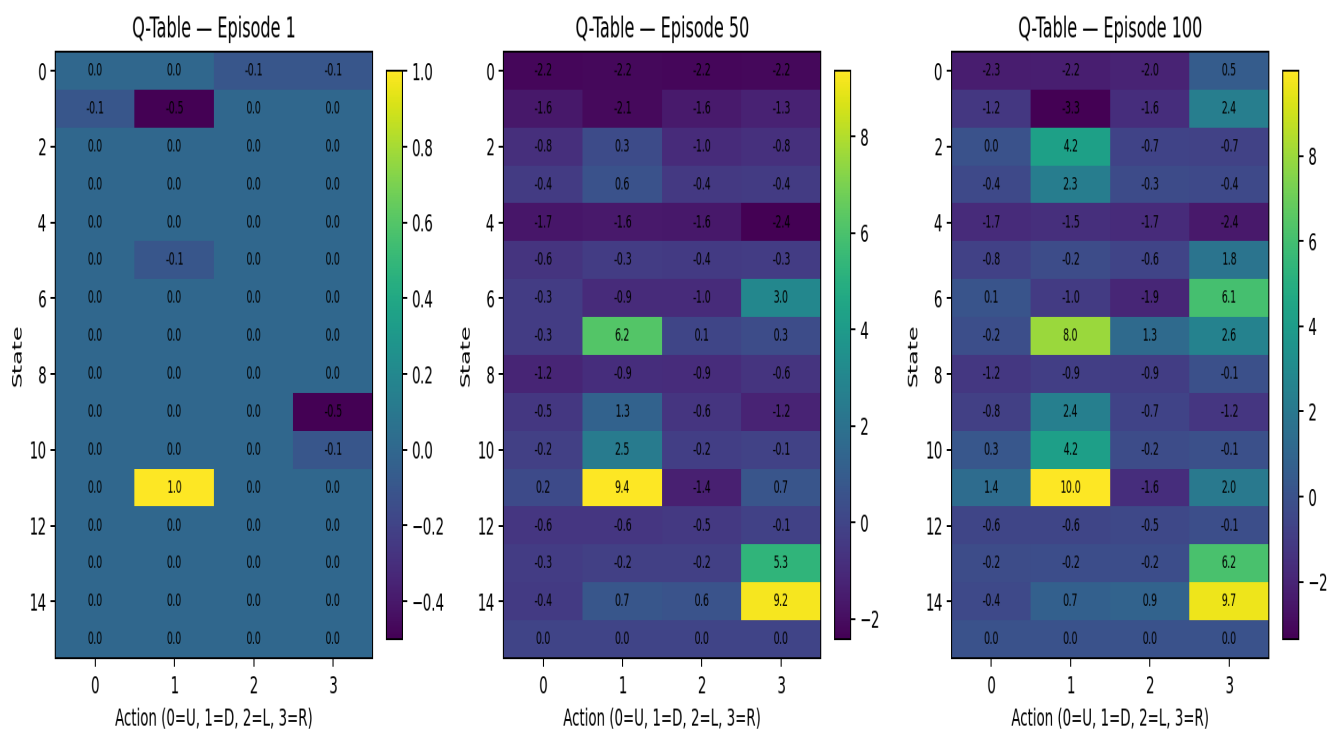
9. Result

The Q-Learning agent learns a goal-directed policy from experience. The experiment demonstrates that repeated reward-based updates can transform an initially zero Q-table into a useful policy.

10. Conclusion

The 4×4 Grid World provides a compact demonstration of Q-Learning. The learned Q-values, final policy and cumulative reward curve collectively show the progression of learning and the effect of exploration and exploitation.

Q-Table Evolution — Episodes 1, 50 and 100



Cumulative Reward per Episode

